THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 7 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Can a triangle have sides 3 cm, 4 cm and 5 cm? 1

(a) Only if it is right-angled **(b)** Only if it is isosceles
(c) No **(d)** Yes

2. Two angles of a triangle are 50° and 60°. The third angle is: 1

(a) 60° **(b)** 70° **(c)** 90° **(d)** 80°

3. A triangle with all three sides equal is called: 1

(a) right-angled **(b)** isosceles **(c)** equilateral **(d)** scalene

4. In a triangle, the two interior opposite angles are 40° and 55°. The exterior angle is: 1

(a) 95° **(b)** 85° **(c)** 105° **(d)** 115°

5. The apex angle of an isosceles triangle is 40°. Each base angle is: 1

(a) 80° **(b)** 60° **(c)** 90° **(d)** 70°

6. The perimeter of a triangle with sides 5 cm, 7 cm and 9 cm (in cm) is: 1

(a) 23 **(b)** 21 **(c)** 22 **(d)** 20

7. Two triangles with all three pairs of sides equal are congruent by: 1

(a) SAS **(b)** AAA **(c)** SSS **(d)** ASA

8. In a right-angled triangle one acute angle is 25°. The other acute angle is: 1

(a) 75° **(b)** 65° **(c)** 85° **(d)** 55°

9. Can a triangle have sides 2 cm, 3 cm and 6 cm? 1

(a) Only if it is right-angled **(b)** No
(c) Only if it is isosceles **(d)** Yes

10. Two angles of a triangle are 35° and 75°. The third angle is: 1

(a) 60° **(b)** 80° **(c)** 70° **(d)** 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Can a triangle be formed with sides 6 cm, 8 cm and 10 cm? Give a reason. 2

12. Two angles of a triangle are 48° and 72°. Find the third angle and name the triangle by its angles. 2

13. In $\triangle ABC$, $\angle A = 45^\circ$ and $\angle B = 60^\circ$. Side BC is extended to D. Find the exterior angle $\angle ACD$. 2

14. Can a triangle be formed with sides 3 cm, 5 cm and 9 cm? Give a reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a triangle are in the ratio $1 : 2 : 3$. Find the angles. 3

16. The apex angle of an isosceles triangle is 30°. Find the base angles. 3

17. An exterior angle of a triangle is 120° and one interior opposite angle is 45°. Find the other interior opposite angle and the adjacent interior angle. 3

OR

The angles of a triangle are in the ratio $2 : 3 : 4$. Find the angles. 3

18. A base angle of an isosceles triangle is 70°. Find the apex angle. 3

OR

An exterior angle of a triangle is 110° and one interior opposite angle is 60°. Find the other interior opposite angle and the adjacent interior angle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A triangle has sides 5 cm, 5 cm and 8 cm. Check that it exists, name it by its sides and find its perimeter. 5

OR

In $\triangle ABC$, $\angle A = 40^\circ$ and $\angle B = 65^\circ$. Find $\angle C$, classify the triangle by its angles and find the exterior angle at C. 5

20. A triangle has sides 6 cm, 8 cm and 10 cm. Check that it exists, name it by its sides and find its perimeter. 5

OR

In $\triangle ABC$, $\angle A = 50^\circ$ and $\angle B = 40^\circ$. Find $\angle C$, classify the triangle by its angles and find the exterior angle at C. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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